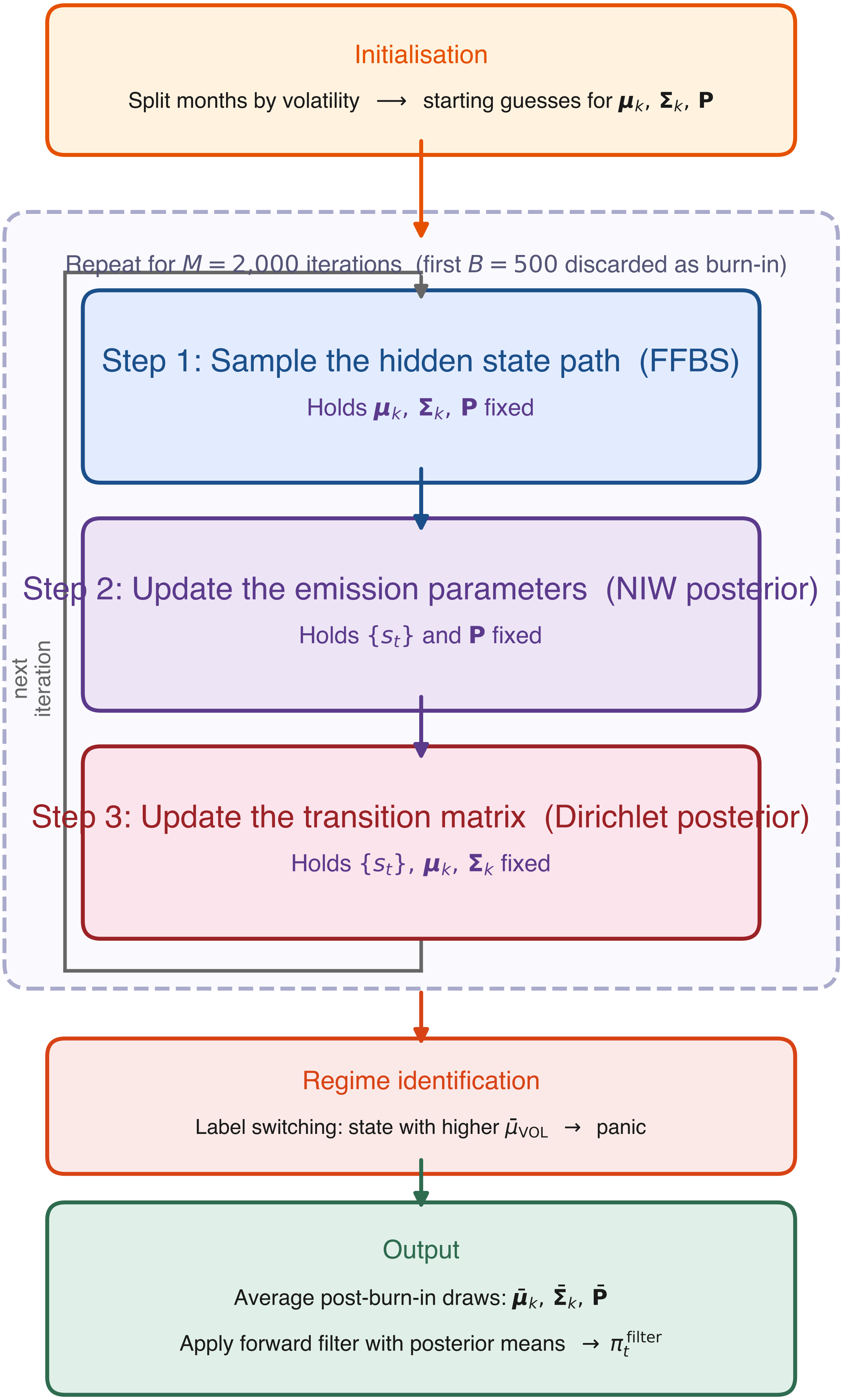
Initialisation
Split months by volatility $\longrightarrow$ starting guesses for $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$, $\mathbf{P}$
Repeat for $M = 2{,}000$ iterations (first $B = 500$ discarded as burn-in)
Step 1: Sample the hidden state path (FFBS)
Holds $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$, $\mathbf{P}$ fixed
Step 2: Update the emission parameters (NIW posterior)
Holds $\{s_t\}$ and $\mathbf{P}$ fixed
Step 3: Update the transition matrix (Dirichlet posterior)
Holds $\{s_t\}$, $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$ fixed
next iteration
Regime identification
Label switching: state with higher $\bar{\mu}_{\mathrm{VOL}}$ $\rightarrow$ panic
Output
Average post-burn-in draws: $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, $\bar{\mathbf{P}}$
Apply forward filter with posterior means $\rightarrow$ π_t^{filter}